Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.36)

Metric

$$ds^2 = -2\,e^{(2\,\rho x)} \mathrm{d}u \otimes \mathrm{d}u + \mathrm{d}u \otimes \mathrm{d}v + \mathrm{d}v \otimes \mathrm{d}u + \mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\,\sqrt{2}e^{(\rho x)}\right)\, du$$

$$n = \left(\sqrt{2}e^{(\rho x)}\right)\, du + \left(-\sqrt{2}e^{(-\rho x)}\right)\, dv$$

$$m = \left(\frac{1}{2}\,\sqrt{2}\right)\, dx + \left(\frac{1}{2}i\,\sqrt{2}\right)\, dy$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}\right)\, dx + \left(-\frac{1}{2}i\,\sqrt{2}\right)\, dy$$

Contravariant components

$$l = \left(\frac{1}{2}\,\sqrt{2}e^{(\rho x)}\right)\, dv$$

$$n = \left(-\sqrt{2}e^{(-\rho x)}\right)\, du + \left(-\sqrt{2}e^{(\rho x)}\right)\, dv$$

$$m = \left(\frac{1}{2}\,\sqrt{2}\right)\, dx + \left(\frac{1}{2}i\,\sqrt{2}\right)\, dy$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}\right)\, dx + \left(-\frac{1}{2}i\,\sqrt{2}\right)\, dy$$

Directional Derivatives

$$D = \frac{1}{2}\,\sqrt{2}e^{(\rho x)}\,\partial_v$$

$$\Delta = -\sqrt{2}e^{(-\rho x)}\,\partial_u - \sqrt{2}e^{(\rho x)}\,\partial_v$$

$$\delta = \frac{1}{2}\,\sqrt{2}\,\partial_x + \frac{1}{2}i\,\sqrt{2}\,\partial_y$$

$$\overline{\delta} = \frac{1}{2}\,\sqrt{2}\,\partial_x - \frac{1}{2}i\,\sqrt{2}\,\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$\lambda = 0$

$\nu = 2\sqrt{2}\rho$

$\alpha = \frac{1}{4}\sqrt{2}\rho$

$\beta = \frac{1}{4}\sqrt{2}\rho$

$\tau = 0$

$\pi = 0$

$\epsilon = 0$

$\gamma = 0$

Ricci Tensor Components

$\Phi_{00} = 0$

$\Phi_{01} = 0$

$\Phi_{02} = 0$

$\Phi_{10} = 0$

$\Phi_{11} = 0$

$\Phi_{12} = 0$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$\Phi_{22} = 4\,\rho^2$

$\Lambda = 0$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$\Psi_2 = 0$

$\Psi_3 = 0$

$\Psi_4 = 4\,\rho^2$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "N"